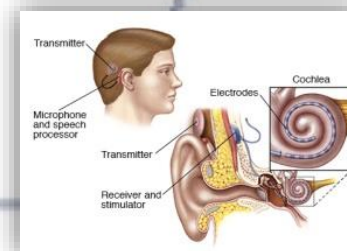
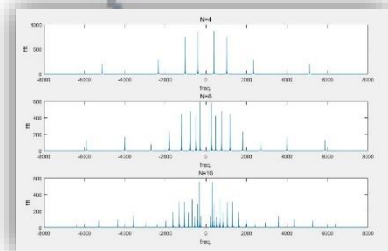
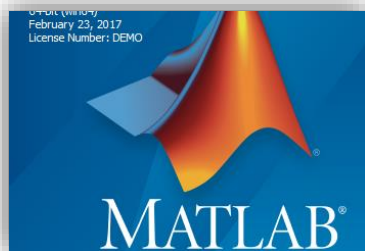


# 信号与系统实验

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## Signals and Systems (Lab)

# Lab 1: MATLAB Programming

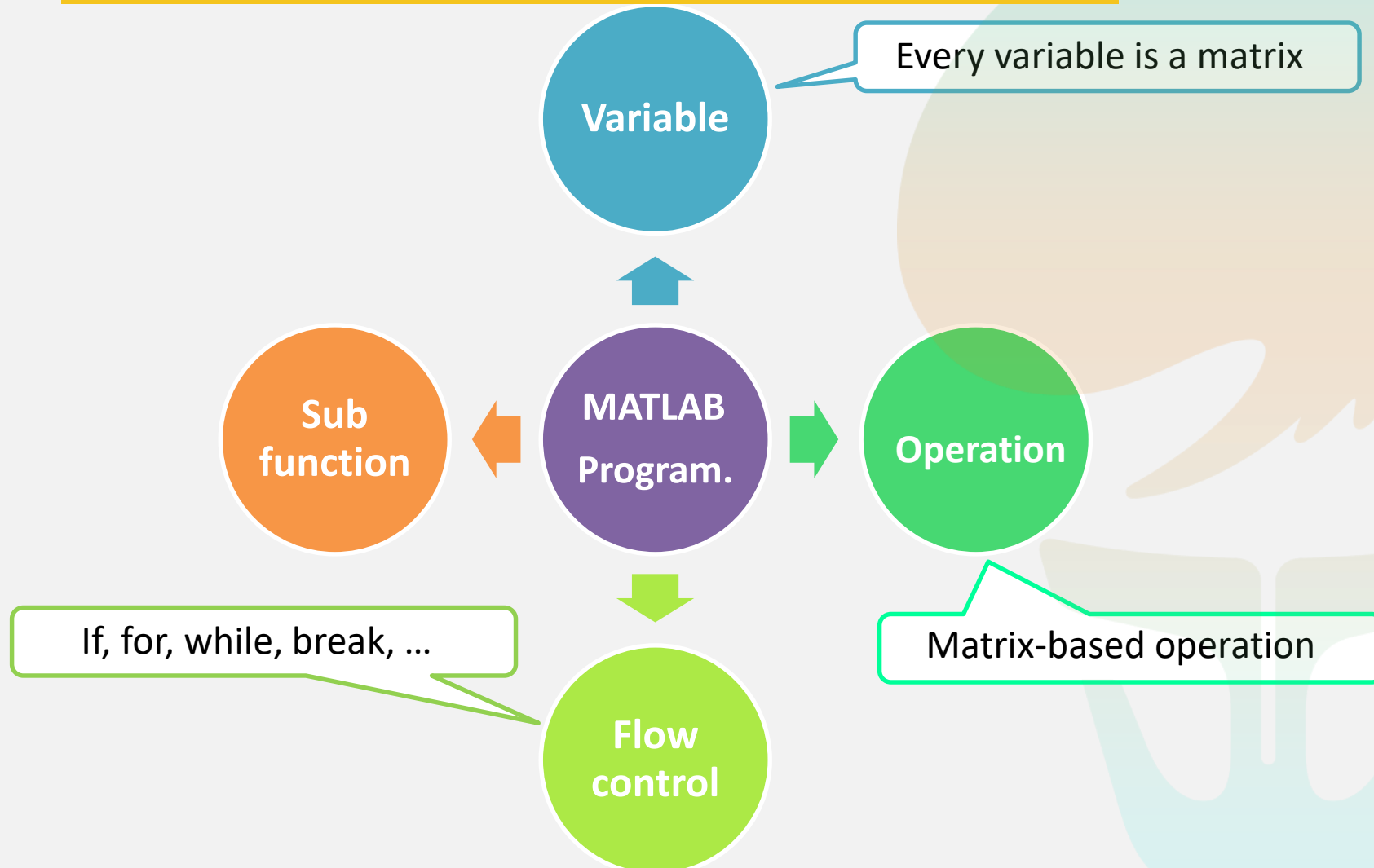
Part2

Dr. **Wu Guang**

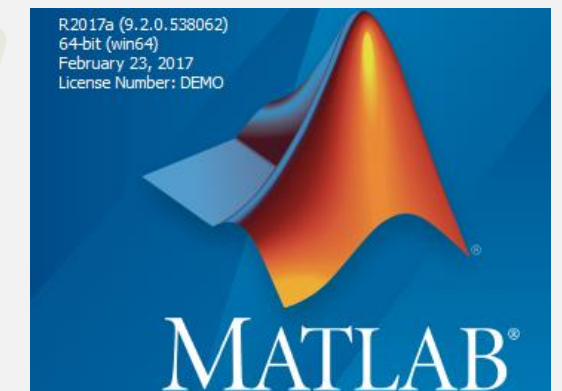
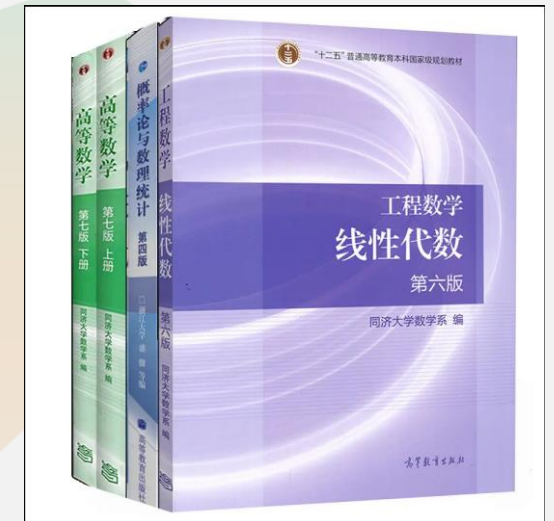
[wug@sustech.edu.cn](mailto:wug@sustech.edu.cn)

**Electrical & Electronic Engineering**  
**Southern University of Science and Technology**

## 7.1 Preliminary knowledge



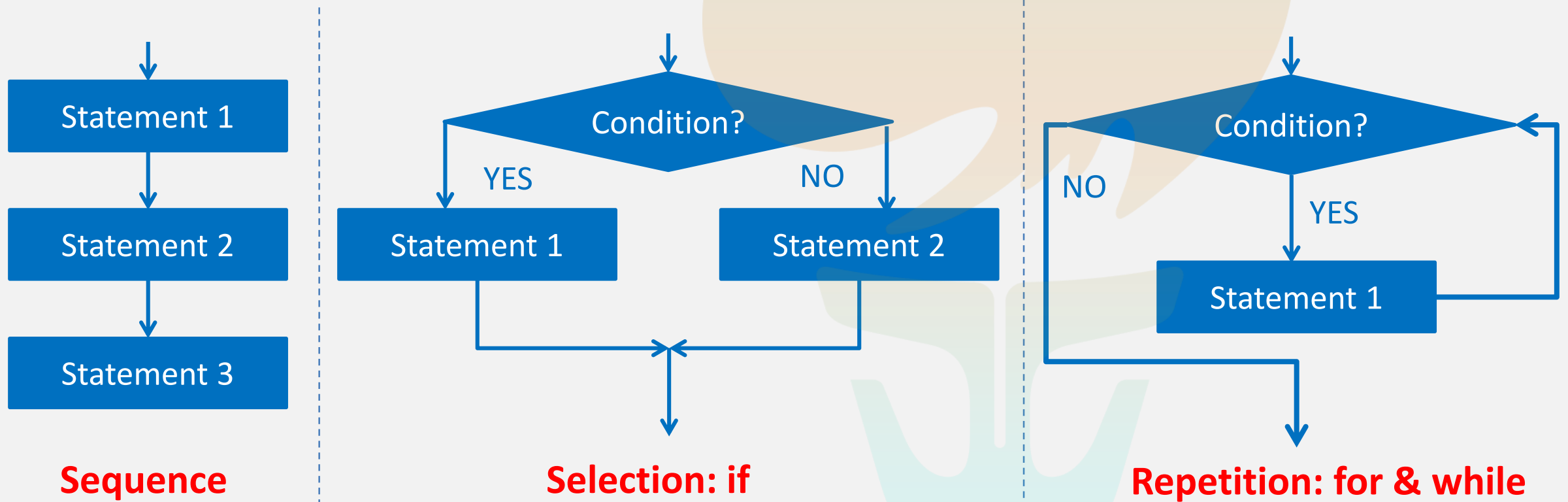
Linear algebra



## 7.2 Flow Control

MATLAB program has three basic structures:

- Sequence, Selection and Repetition



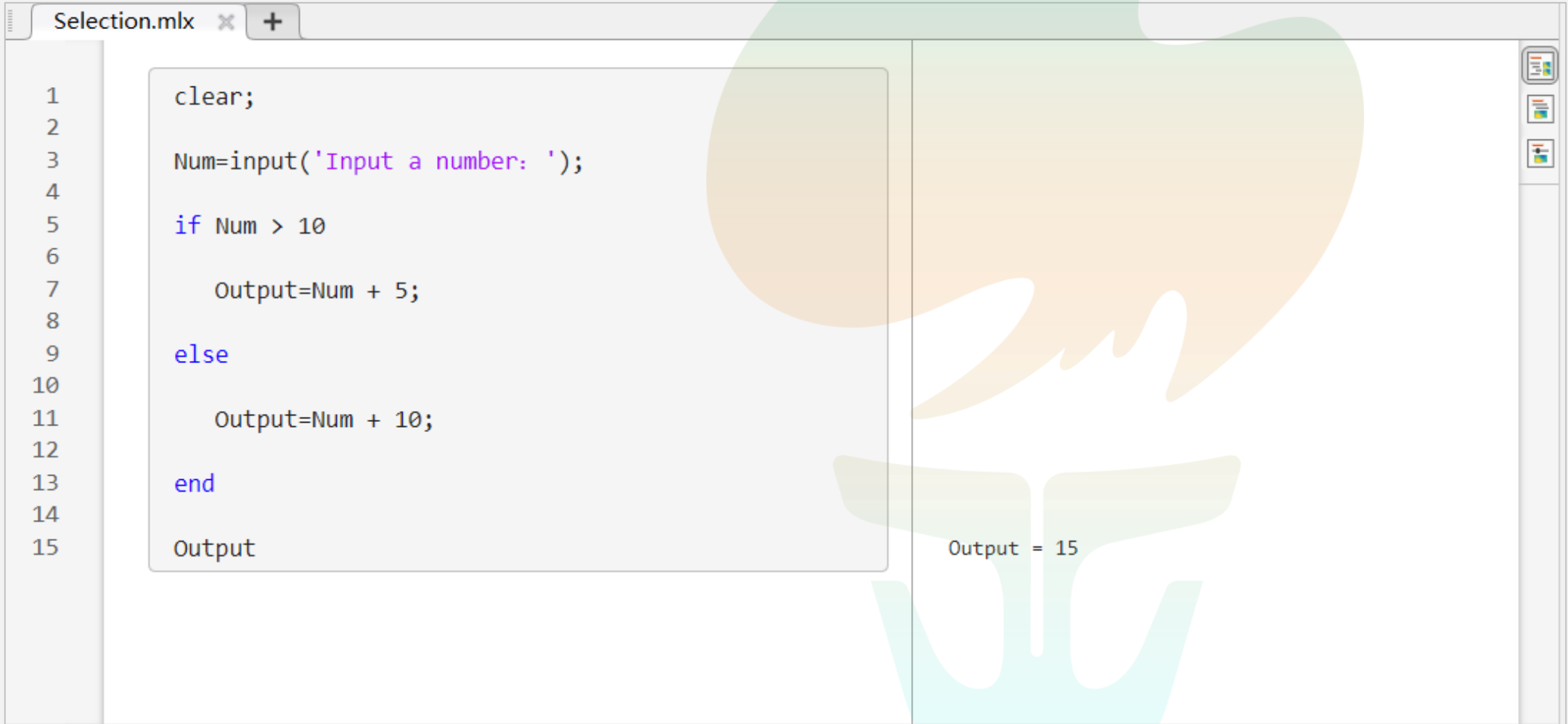
## 7.3 Exercise 1.1 Selector.m

---



**Write a script to add '5' to a number if it is greater than '10', else add '10'.**

## 7.3 Exercise 1.1 Selector.m



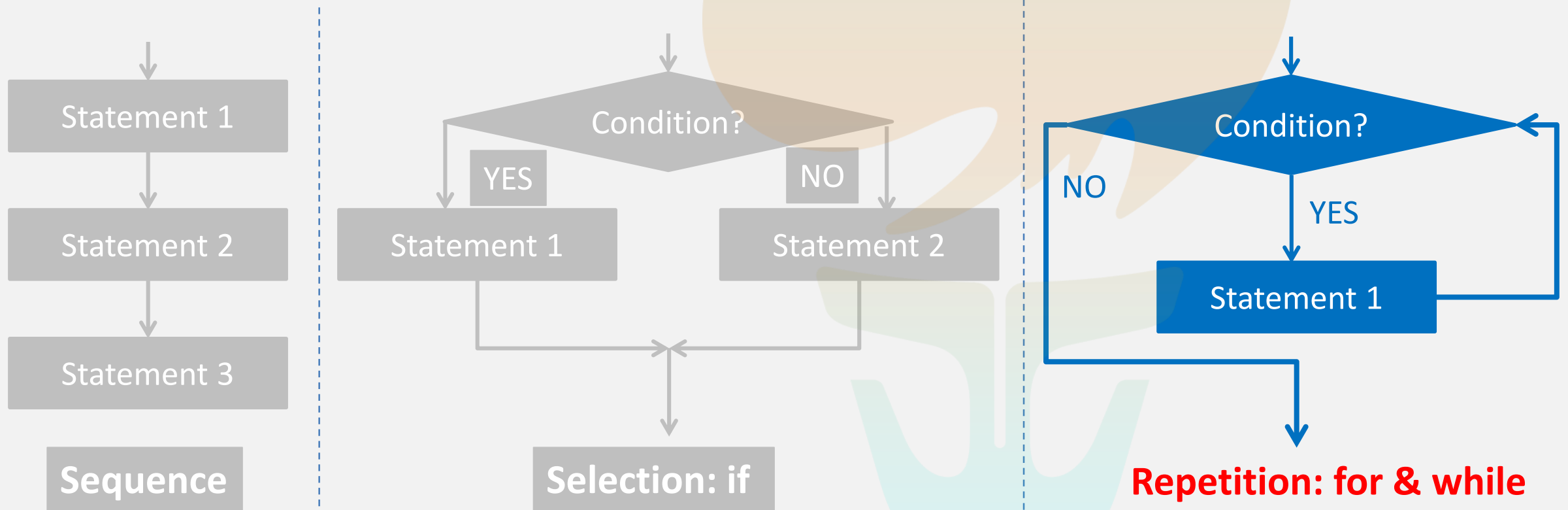
```
1 clear;
2
3 Num=input('Input a number: ');
4
5 if Num > 10
6     Output=Num + 5;
7
8 else
9     Output=Num + 10;
10
11 end
12
13 Output
```

Output = 15

## 7.4 Loop-Structure

Similar to almost all programming language, MATLAB program has three basic structures:

- Sequence, Selection and Repetition



## 7.5 Loop-Structure

---

```
for i=Index_Array
```

```
    MATLAB Commands
```

```
end
```

```
while (condition)
```

```
    MATLAB Commands
```

```
end
```



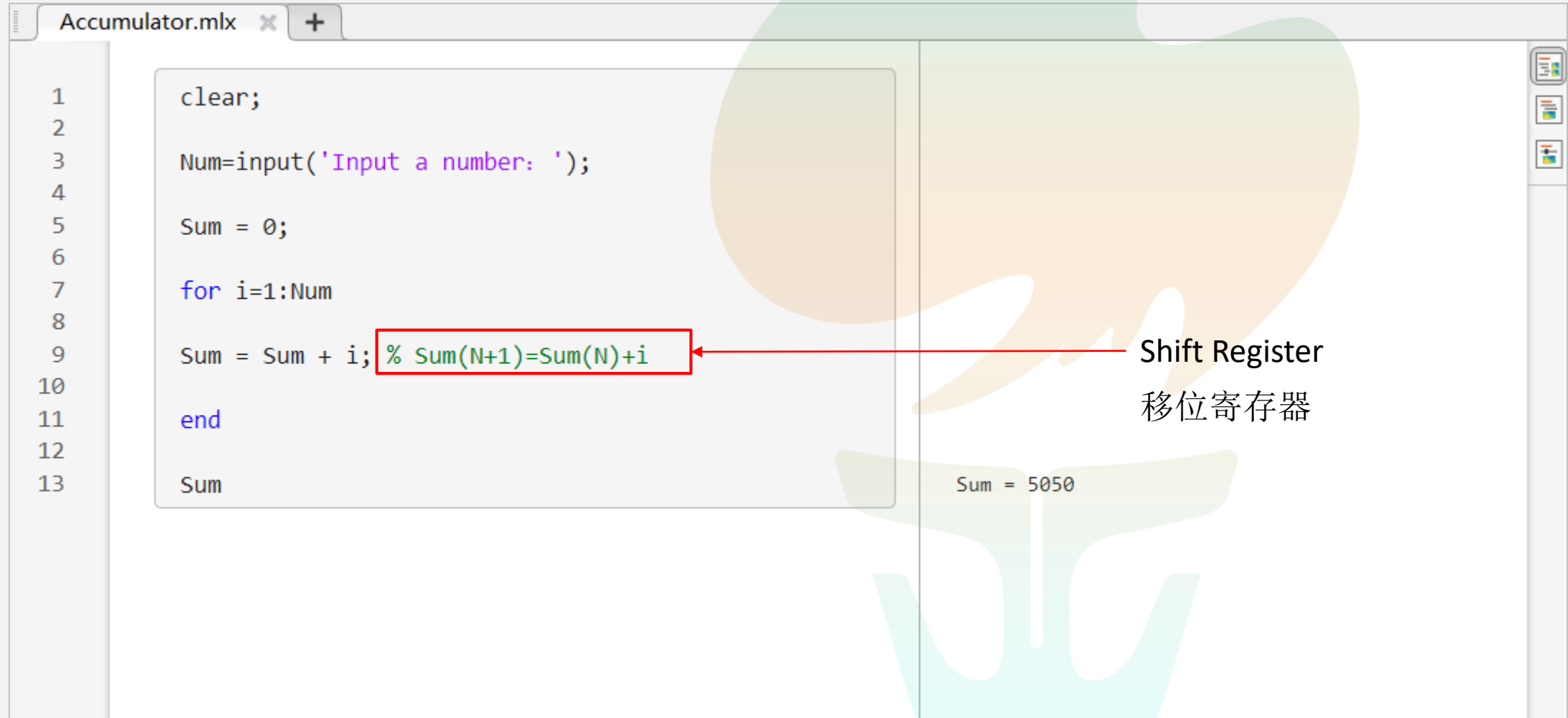
## 7.5 Exercise 1.2 Accumulator.m

---



**Write a script to output the sum of numbers from 1 to the number input.**

## 7.5 Exercise 1.2 Accumulator.m



```
1 clear;
2
3 Num=input('Input a number: ');
4
5 Sum = 0;
6
7 for i=1:Num
8
9   Sum = Sum + i; % Sum(N+1)=Sum(N)+i
10
11 end
12
13 Sum
```

Shift Register  
移位寄存器

Sum = 5050

## 7.5 Exercise 1.2 Accumulator\_while.m



```
1 clear;
2
3 Num=input('Input a number: ');
4
5 i=0;
6
7 Sum = 0;
8
9 while (1)
10
11     if i>Num, break; end
12
13     Sum = Sum + i; % Sum(N+1)=Sum(N)+i
14
15     i=i+1;
16
17 end
18
19 Sum
```

Stop Condition

Sum = 5050

## 7.6 Operators (Logical)

1. == Equal to
2. ~= Not equal to
3. < Strictly smaller
4. > Strictly greater
5. <= Smaller than or equal to
6. >= Greater than equal to
7. & And operator
8. | Or operator
9. ~ Not operator

```
if ((a>3) & (b==5))  
    Some MATLAB Commands;  
end
```

```
if (a<3)  
    Some MATLAB Commands;  
elseif (b~=5)  
    Some MATLAB Commands;  
end
```

```
if (a<3)  
    Some MATLAB Commands;  
else  
    Some MATLAB Commands;  
end
```

## 7.7 Writing User Defined Functions

- Functions are m-files which can be executed by specifying some inputs and supply some desired outputs.
- The code telling the MATLAB that an m-file is actually a function is

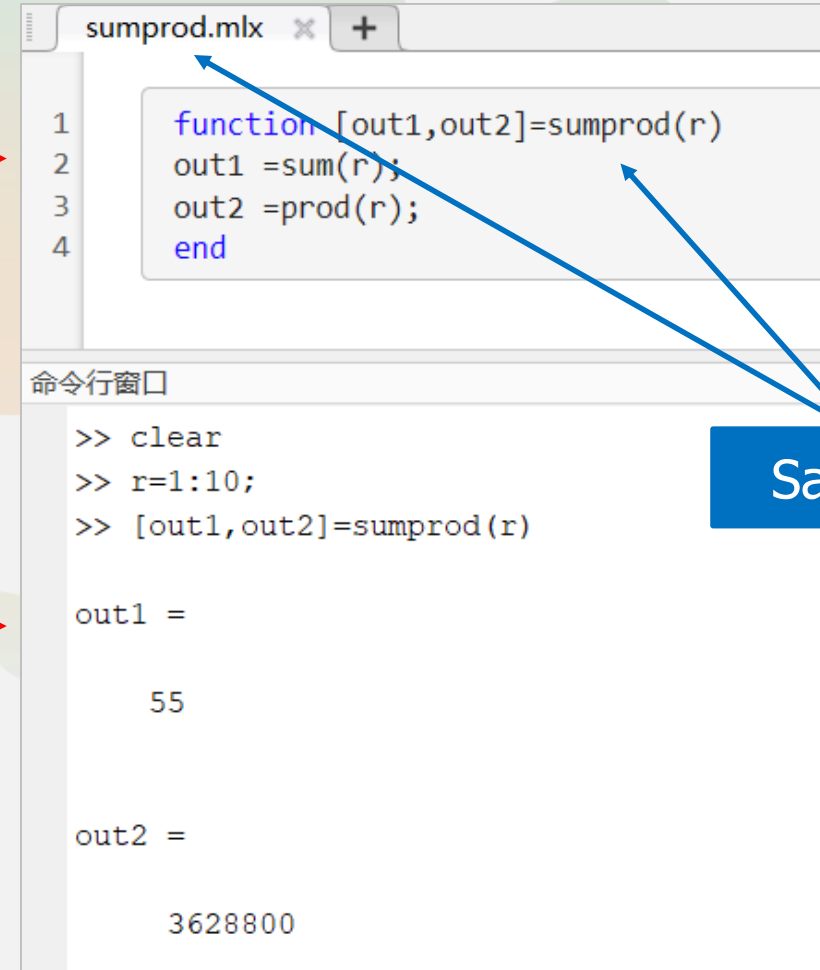
```
function out1=functionname(in1)
function out1=functionname(in1,in2,in3)
function [out1,out2]=functionname(in1,in2)
```

- You should write this command **at the beginning of the m-file** and you should save the m-file **with a file name same as the function name**

## 7.7 Writing User Defined Functions

For example, the function takes an input array and returns the sum and product of its elements as outputs.

The function `sumprod(.)` can be called from command window.



The image shows a MATLAB interface with two windows. The top window, titled 'sumprod.mlx', contains a function definition:

```
1 function [out1,out2]=sumprod(r)
2 out1 =sum(r);
3 out2 =prod(r);
4 end
```

The bottom window, titled '命令行窗口' (Command Window), shows the execution of the function:

```
>> clear
>> r=1:10;
>> [out1,out2]=sumprod(r)

out1 =

    55

out2 =

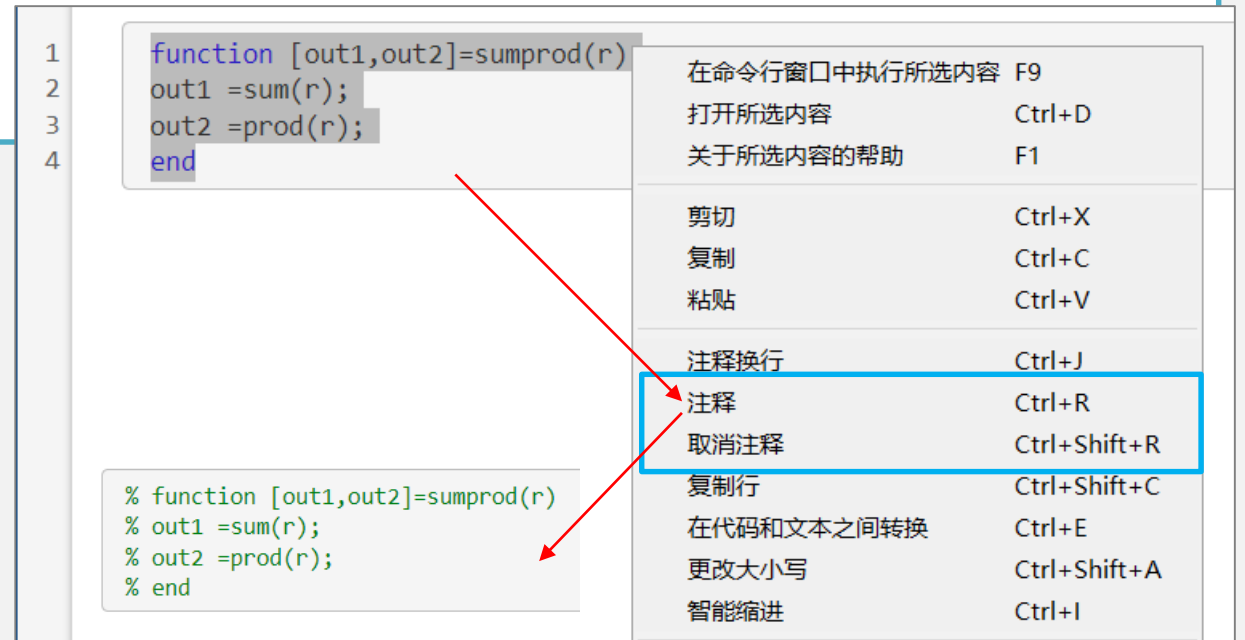
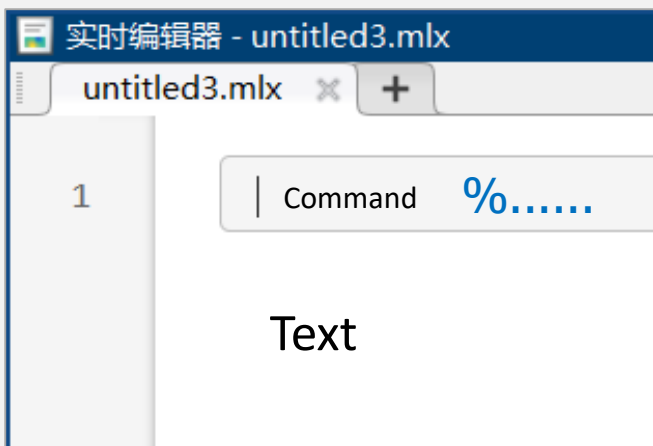
 3628800
```

Red arrows point from the text boxes to the function definition and the command window. Blue arrows point from the function name 'sumprod' in the code to the function call 'sumprod(r)' in the command window. A blue box labeled 'Same Name' is positioned near these arrows.

Same Name

## 7.8 Tip1: Comment

- “%” is the neglect sign for MATLAB (equivalent of “//” in C). Anything after it on the same line is neglected by MATLAB compiler.
- For example, sometimes slowing down the execution is done deliberately for observation purposes. You can use the command “pause” for this purpose
- pause %wait until any key
- pause(3) %wait 3 seconds



```
1 function [out1,out2]=sumprod(r)
2 out1 =sum(r);
3 out2 =prod(r);
4 end
```

% function [out1,out2]=sumprod(r)  
% out1 =sum(r);  
% out2 =prod(r);  
% end

|               |              |
|---------------|--------------|
| 在命令行窗口中执行所选内容 | F9           |
| 打开所选内容        | Ctrl+D       |
| 关于所选内容的帮助     | F1           |
| 剪切            | Ctrl+X       |
| 复制            | Ctrl+C       |
| 粘贴            | Ctrl+V       |
| 注释换行          | Ctrl+J       |
| 注释            | Ctrl+R       |
| 取消注释          | Ctrl+Shift+R |
| 复制行           | Ctrl+Shift+C |
| 在代码和文本之间转换    | Ctrl+E       |
| 更改大小写         | Ctrl+Shift+A |
| 智能缩进          | Ctrl+I       |

## 7.8 Tip2: Try to Avoid Loops

- Instead of loops, try to use matrix operators and built-in functions.
- Example: let  $a = [a_1 \ a_2 \ a_3]$  and  $b = [b_1 \ b_2 \ b_3]$ , write a function to calculate

$c = [a_1+b_1 \ a_1+b_2 \ a_1+b_3; a_2+b_1 \ a_2+b_2 \ a_2+b_3; a_3+b_1 \ a_3+b_2 \ a_3+b_3]$

### • Solution 1

```
function c = add1(a,b)
c = zeros(3,3);
for m=1:3
    for n=1:3
        c(m,n) = a(m) + b(n);
    end;
end;
```

### Solution 2

```
function c = add2(a,b)
c = diag(a) * ones(3) + ones(3) * diag(b);
```

built-in function

内联函数



## 7.9 Lab Assignments

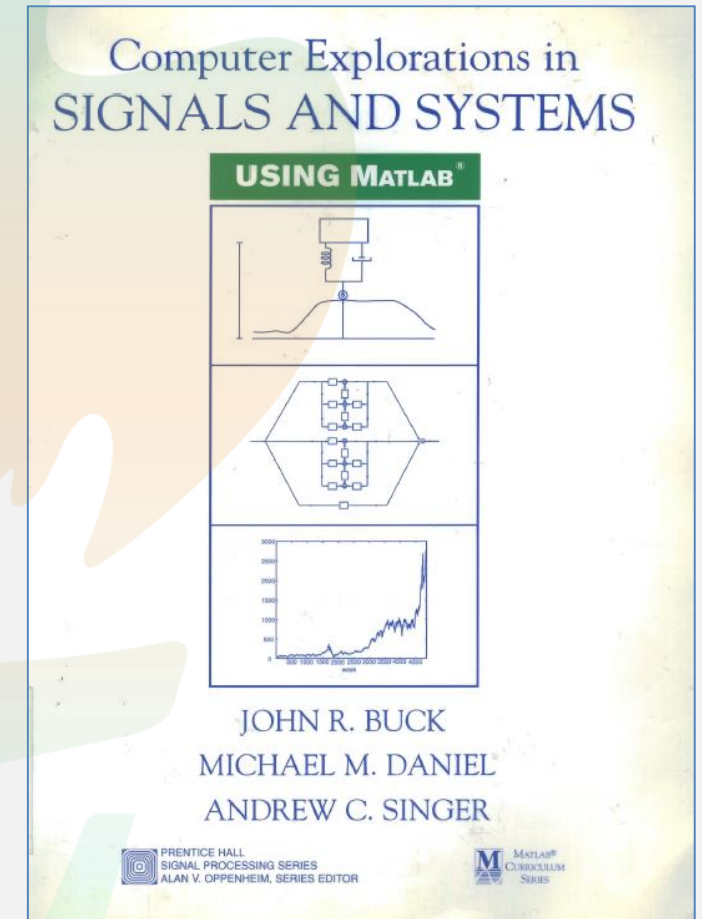
### ■ 1.4 Properties of Discrete-Time Systems

Discrete-time systems are often characterized in terms of a number of properties such as linearity, time invariance, stability, causality, and invertibility. It is important to understand how to demonstrate when a system does or does not satisfy a given property. MATLAB can be used to construct counter-examples demonstrating that certain properties are not satisfied. In this exercise, you will obtain practice using MATLAB to construct such counter-examples for a variety of systems and properties.

#### Basic Problems

For these problems, you are told which property a given system does not satisfy, and the input sequence or sequences that demonstrate clearly how the system violates the property. For each system, define MATLAB vectors representing the input(s) and output(s). Then, make plots of these signals, and construct a well reasoned argument explaining how these figures demonstrate that the system fails to satisfy the property in question.

- The system  $y[n] = \sin((\pi/2)x[n])$  is not linear. Use the signals  $x_1[n] = \delta[n]$  and  $x_2[n] = 2\delta[n]$  to demonstrate how the system violates linearity.
- The system  $y[n] = x[n] + x[n + 1]$  is not causal. Use the signal  $x[n] = u[n]$  to demonstrate this. Define the MATLAB vectors  $x$  and  $y$  to represent the input on the interval  $-5 \leq n \leq 9$ , and the output on the interval  $-6 \leq n \leq 9$ , respectively.



# Question ?

---

